

Python Programming – Mock Paper Part 1

Candidates may use **VS Code, or Jupyter Notebook** as the programming platform.

All programs must be written using **clean, well-structured, and meaningful code**, with appropriate variable names, proper indentation, and clear logic.

Submissions must be in **.ipynb** format.

All required files must be uploaded to the **Learning Management System (LMS)** within the specified deadline.

1. Variables & Built-in Functions

- a. Use `input()` to collect a user's `full_name`, `city`, and `hobby`.
- b. Ask the user for their `birth_year`, convert it to an **integer**, and store it.
- c. Calculate the user's age by subtracting the `birth_year` from 2026 and print the result.
- d. Use the `len()` function to print the number of characters in the `full_name` variable.

2. Data Types

- a. Declare a variable `is_student` with a **boolean** value.
- b. Create a variable `hourly_rate` with a **float** value and print it using the `type()` function.
- c. Use the `str()` function to convert an integer `100` into a string.
- d. Explain in a comment the difference between a **dynamically typed** language (like Python) and a statically typed language.

3. Lists & Tuples

- a. Create a list named `colors` with 5 different color names.
- b. Update the 2nd item in the list to "Purple" and remove the last item.
- c. Create a **tuple** named `dimensions` containing three numeric values.

- d. Demonstrate what happens if you try to change a value in the `dimensions` tuple (use a comment or code).

4. Strings

- a. Concatenate the strings "Python", "is", and "Powerful" with spaces between them.
- b. Use the `find()` method to locate the word "fun" in the string "Coding is fun and creative".
- c. Slice the first 6 characters from the string "SoftwareEngineering".
- d. Use a string method to check if the string "Admin_Panel" starts with the word "Admin".

5. Loops

- a. Write a `for` loop to print the square of every number from 1 to 5.
- b. Use a `for` loop to iterate through the list `['DevOps', 'Cloud', 'Data']` and print each item.
- c. Create a `while` loop that prints a counter starting at 10 and stops when it reaches 1.
- d. Write a loop that prints the following pattern:

```
*
**
***
****
*****
```

6. Conditionals

Create a Python script that evaluates an employee's performance based on their annual KPI score.

- a. **Input Gathering:** Write a script that prompts the user to enter the **Employee Name** and their **KPI Score** (as an integer).
- b. **Validation Check:** Use an `if` statement to ensure the score is within the valid range of **0 to 100**. If the score is invalid, print an error message.
- c. **Logic Implementation:** Use `elif` statements to determine the performance level based on these ranges:
 - **90 – 100:** Exceptional

- **75 – 89:** Commendable
- **60 – 74:** Satisfactory
- **Below 60:** Needs Review

d. F-String Formatting: Display the final result using an **f-string** to create a personalized message in the following format: Employee: [Name] | Score: [Score] | Status: [Level].

7. Functions

- Write a function named `greet_user` that takes a name as a parameter and prints "Hello [name]".
- Create a function `multiply_numbers` that returns the product of two parameters.
- Define a function that calculates the area of a circle ($\text{Area} = 3.1416 \times r^2$).
- Explain the difference between a function **parameter** and an **argument** in a comment.